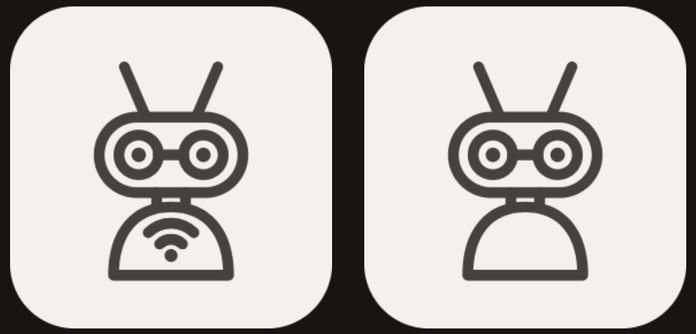
LabaLab
Language, Affect and Behaviour in AI

ROBOT GUIDE

Reachy Mini



Switch it on, connect it, and use every screen and button: talking, group conversations and Wizard of Oz.

LabaLab · BGU · Room –132

MEET REACHY

Two kinds of Reachy in the lab



Reachy Mini

The wireless one. Each has its name on it: reachy1, reachy2 …

Connects: WiFi

Power: its own battery

Runs: the lab’s Reachy app

Main button: Talk



Reachy Mini Lite

The cabled one. It has no WiFi and no battery.

Connects: USB cable

Power: wall power supply

Runs: Pollen’s Reachy Mini app

Main button: Open

Both talk and listen, move their head and antennas, and see through a camera in the head.

ONCE PER LAPTOP

Install two apps, add one key

1

LabaLab Robot Hub

Finds the robots and starts the Reachy app. Windows, no admin needed. In PowerShell, paste the line below and press Enter.

2

Reachy Mini app

Pollen's own app. It runs the Lite and updates the robots' software. Download it from:

hf.co/reachy-mini/#/download

3

A key

Ask lab staff for your own key. In Robot Hub: paste it, press **Test and save**, wait for **It works**.

Reachy talks with a **Gemini** or **GPT** key.

```
irm https://github.com/labbgulab-lab/robot-hub/releases/latest/download/install.ps1 | iex
```

On GitHub: **github.com/labbgulab-lab/robot-hub**. Line blocked? Releases › download the zip › **Install Robot Hub.cmd**.

EVERY SESSION

Switch on and connect

1. **Charge it fully**
   It can’t show its battery. Flat means it stops mid-sentence.
2. **Laptop on HOT-WiFi**
   Or your phone hotspot, if the robot knows it. VPN off.
3. **Switch the robot on**
   Give it about a minute to join the WiFi.
4. **Open LabaLab Robot Hub**
   After about 30 s its card shows **Available**.
5. **Press Connect**
   The robot says a sentence out loud: the link works.
6. **Press Talk**
   The Reachy app opens in its own window.

Shortcuts

Auto-connect at the top connects every robot the moment it shows up.

Scan looks for robots again.

Closing the window does not stop the robots. To stop everything: **Settings › About › Quit**.

ROBOT HUB

Reachy’s card, part by part

On the card	What it means or does
Available	Found on the WiFi, not linked yet.
Connect	Links the robot to this laptop. It says a sentence out loud to prove it.
Connected	Linked and ready. **Disconnect** lets it go again.
Talk	Opens the Reachy app on its Talk tab.
Speaks through	Which of your keys this robot uses. Default: your first working key.
… (details)	Its address, software version and last seen. **Copy details** for a bug report.
Lost	The link dropped. Press **Connect now**. If it stays, press **See log**.
In use	Another laptop is using it. Ask that person, or lab staff.

THE REACHY APP

Three tabs, three ways to use Reachy

Talk

One Reachy has a conversation with the people in front of it.

Group

Two or more Reachys debate or chat with each other and the room.

Wizard of Oz

You drive it: type a line or fire a ready cue, and Reachy says it.

Along the top

Robots · back to the robot list in Robot Hub

Saving transcript · this conversation is being saved

Which Reachy? · pick the robot for this tab

Logs · what Robot Hub is doing, live

TALK TAB · 1 OF 3

Robot and How it talks

Setting or button	What it does
Details	Everything Robot Hub knows about this robot.
Stop / Start system	Shuts the robot’s side down, or brings it back up (about 7 to 10 s).
Brain	The AI that listens and thinks in the next conversation.
Language	**Auto** answers in the language the person spoke. Or always Hebrew, or always English.
Brain voice	The brain’s own voice. It wins over the character’s voice.
ElevenLabs v4 voice	Speak with an ElevenLabs voice instead (pick it in **EL voice**). The brain still thinks.
Camera vision	Looks through its camera when someone asks what it sees. Never comments unasked.

Changes here apply to the **next** conversation, not the one running.

TALK TAB · 2 OF 3

Give Reachy a character

1. **Pick one in Start from**
   A ready-made character. Picking it saves it and switches it on.
2. **Or write your own in the box**
   Up to 1500 characters. It adds to Reachy's base persona.
3. **Pick its Voice**
   Used while **Brain voice** is set to Default.
4. **Press Save**
   Saves the text and voice, and switches the character on.

The buttons

Character switch: off means the plain base persona. Off keeps your text.

Save: stores text and voice, switches it on.

Clear: wipes the text and goes back to the base persona.

Example text: "You are a cheerful museum guide. Keep every answer to two sentences."

TALK TAB · 3 OF 3

Run a conversation

1. **Check Brain, Language and Character**
   They are used from the moment you press Start.
2. **Press Start conversation**
   Reachy speaks first.
3. **Talk to it, facing the robot**
   One person at a time works best.
4. **End it: say “goodbye” or “להתראות”**
   Or press **End conversation**.

While it runs

Turn counts the exchanges. **This turn** times the current one.

Event log shows everything that happens, live.

Copy logs copies it all, for a bug report.

Saving transcript at the top means the conversation is being saved on this laptop.

GROUP TAB · 1 OF 3

Run a group conversation

1. **Connect two or more Reachys**
   Plug the DJI Mic Mini receiver into the laptop; the speaker wears the mic.
2. **Open the Group tab**
   In Robot Hub, press Group talk in the Reachy app band.
3. **Tick Included on each robot**
4. **Write a Topic, pick Language and Minutes**
5. **Press Get ready**
   Connects the robots and tests every service, before people arrive.
6. **Press a mode, then Stop to end**
   Robots say a closing line. **Stop now** ends at once.

The mode buttons

Group talk: robots debate the topic; anyone can jump in.

Interrupt: a person talks, robots cut in to argue, cheer or correct.

Free talk: no topic, chat about anything.

+1 min: more time for this conversation.

Disconnect: releases the robots.

GROUP TAB · 2 OF 3

Each robot’s card

On the card	What it does
Included	Takes part in the conversation. Untick to leave this robot out.
Listens	Its own mic hears the people when the wireless mic isn’t in use.
Personality	Its side on the topic and how it argues. Applies from its next line.
Voice · Writer	The voice it speaks with, and the AI that writes its lines. Give each robot its own voice.
Interrupting 0–5	How often it cuts in during Interrupt. 0 = never.
Gestures	**Lively**: moves on almost every line. **Calm**: on about half.
Volume	Its speaker. Above 100% the voice itself is boosted.
Copy to others	Copies the settings you choose to other robots.
‹ ›	Moves it one seat left or right, as the audience sees them.

GROUP TAB · 3 OF 3

The conversation and the room

This conversation

Topic: what they debate. Empty: they pick one.

Language: Auto, Hebrew or English.

Minutes: 0.5 to 60. Then they say goodbye.

Lab intro: each robot first says a line about the lab.

While it runs

Steer: type an idea, press Enter. The next line follows it.

Keys 0 to 5: set every robot’s interrupting at once.

Choir: all robots speak as one voice.

Room and network

Gap: silence between robot lines. Longer leaves people room to jump in.

Voice threshold: how loud a voice must be to take the floor.

People cut in: Off, Careful, Normal or Eager.

Audience mic: Auto uses the wireless mic on the laptop, else the robots’ mics.

Slow network: on for a phone hotspot.

Heads turn the other way: on if robots look away from the speaker.

Simulate on this laptop: rehearse with no robots.

WIZARD OF OZ TAB

You type, Reachy says it

Say anything

1. Type a line in the box. Reachy records it while you type.

2. Press **Say**. It speaks in its own voice, then goes back to breathing.

3. STOP or Esc silences it at once. Safe any time.

Quick saves S, D, F: three ready lines on those keys. **Edit** changes them.

Said: everything it said, in order.

Needs a Gemini key on the robot, or you see NO_KEY.

Audio cues board

Ready-made lines and moves for a show. Firing one cuts into a running conversation.

Space: fires the outlined cue and moves on.

↑ ↓: move without firing. **Letter keys:** fire a cue directly.

Edit a cue: its name, what Reachy says (up to 600 characters), its hotkey and a motion: emotion, dance or head.

Reload script · **Rebuild audio** after changing lines.

REACHY MINI LITE

The USB Reachy

1. **Install the Reachy Mini app, once**
   From **hf.co/reachy-mini/#/download**, or **Set up** on the card.
2. **Plug in power, then USB**
   Power supply into the wall, USB cable into the laptop.
3. **Open the Reachy Mini app, keep it running**
   It is the Lite's brain, and it updates the robot's software.
4. **In Robot Hub, press Connect**
   The card lights up a few seconds after you plug it in.
5. **Press Open, then Play on Conversation**
   To stop, say "go to sleep", or stop it in the app.

⚠ Watch out

No WiFi at all. It never shows up on a network, only over USB.

Unplug it and the app closes by itself about 30 s later.

The app holds one robot. Keep it on the Lite. A wireless Reachy can run at the same time.

WHEN SOMETHING GOES WRONG

What you see, and what to do

What you see	What to do
Not nearby	Is the laptop on the robot’s WiFi? Switch the robot off and on, then press Scan.
Lost	Press Connect now. If it stays, press See log.
In use / Update needed	Another laptop has it: ask that person. Too old: ask lab staff to update it.
Moves, never answers	Is Speaks through a Gemini or GPT key? Speak close to the robot: it hears with its own mic.
Voice stutters	ElevenLabs v4 voice stutters with GPT-Live-1. Use it with Gemini, or switch it off.
Two robots swap places	They have the same name. Ask lab staff to rename one.
NO_KEY in Wizard of Oz	Typed lines need a Gemini key on the robot.
Stops mid-sentence	The battery is flat. Charge it fully.
First Talk is slow	It is installing its app on the robot. Wait a few minutes.
Nothing helps	Logs › Save a report for the lab. Send it to lab staff.

REMEMBER

Four habits for a smooth session

Charge fully first

Reachy can’t warn you. A flat robot drops off mid-sentence.

Same WiFi as the robot

Laptop and robot on HOT-WiFi or the same hotspot. VPN off.

End cleanly

End the conversation, Disconnect, then Settings › About › Quit.

Stuck? Send a report

Logs › Save a report for the lab. Keys are removed for you.